Homework #4

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1 QUESTION 1

1.1 ChatGPT – Not Just an AI…Brainstorming

For this task, I asked ChatGPT to brainstorm ideas for a fitness app. In doing so, it offloaded the cognitive role of generating ideas for a product name for my app. Instead of me performing this action entirely, the interface acts as a conversational partner and based on its replies, it guides me by asking contextual questions to get a better idea of what kind of name suggestions it should come up with. Then, with my guidance via the response to those questions, it uses that context to narrow the process down even further. With each iteration, I gained clarity of my own mental model of what the product is and decide from the list of names that were formulated because I had a clear vision of what the product should be. Not only was ChatGPT performing the cognitive role of idea generation, but it also performed roles that assisted in those cognitive roles, guiding and structuring the idea generation process, distributing associated cognitive tasks throughout the system.

1.2 ChatGPT – Not Just an AI…Brainstorming

For this task, I asked ChatGPT to give me feedback on my app idea and whether it's feasible and if there are other tools like it on the market in order to get feedback on how to be unique compared to my competitors. In doing so, it offloaded the cognitive role of analyzing my app idea and what my competition is as well as evaluating the app idea and how I can be unique amongst my competition. Instead of doing this on my own, I utilized the system to perform this analysis and evaluation and present it in a digestible way for me to analyze and evaluate further. I basically outsourced half of the analysis and evaluation that I would need to do on my own onto the interface from which I could then use to take my next decision.

1.3 ChatGPT – Not Just an AI…Logo Design

Based on a summary of what the app's vision is and goals for users are, I asked ChatGPT to design the logo for the application. The cognitive role I've offloaded onto it is the role of translating the thoughts in my mind into a visual representation. This allowed me to abstract the cognitive and physical processes of translating an idea into an image onto ChatGPT allowing me to focus on the cognitive role of evaluating the output of that translation. In this sense, I've distributed the translation portion onto the interface so that I don't need to and can focus on evaluating whether the output satisfies my needs or not. If it doesn't, iterate over the process until it does. Not only do I no longer need to maintain the mental representation of the logo in my working memory, but I've distributed the load of developing it onto the interface.

1.4 Insights – We're Not Alone in This!

Viewing generative AI as a broader system playing cognitive roles is insightful because it allowed me to view it as an extension of my abilities rather than a tool. Offloading parts of the cognitive tasks of brainstorming, getting feedback, and translating representations demonstrated that I could not only direct my cognitive load onto other areas, but that I could *trust* the interface to perform these tasks and reason with me. Another example of this is the fact that I've distributed searching my browser with ChatGPT. Instead of searching for something on the internet and finding the answer on a webpage, I'll ask ChatGPT what the answer is instead. It's distributed the tasks of retrieving information, analyzing it, and displaying it in an easily digestible way in a single interaction. Instead of me having to do those things, I've offloaded these components onto ChatGPT and all I need do is ingest the information. Overall, viewing generative AI as part of a broader system has allowed me to view it as a partner or extension of myself that shares tasks.

2 QUESTION 2

2.1 Wearable Fitness Trackers and Monitoring Devices – Oh the Politics!

The area I've selected where political motivations determine the design of technology, I encounter regularly is wearable fitness trackers and health monitoring devices. For centuries, human health has been a pinnacle of attention because let's face it – the less health we have, the harder it becomes to accomplish many of our goals. With the rise in technology, we have seen a major shift in how it can be used to assist in monitoring and preventing health risks through tracking all kinds of data such as sleep, daily expenditure, heart rate, and so much more that wouldn't have been possible before. While they might appear to be primarily designed for the benefits of the users, their design is heavily influenced by the competing stakeholders at play.

2.2 The Stakeholders

The stakeholders in this area include the device users, healthcare providers, and the technology companies who create these devices. The users are motivated by the ability to monitor their health status in real time, improve their health, and health risk prevention. Healthcare providers are motivated by accurate data for patient care, predicting health risks, and lowering healthcare costs. Technology companies are motivated by the data they collect to improve UX, enabling improved health and productivity for their users, and monetary profits.

2.3 Ulterior Motives…

The motivation for the device users is specifically affecting the design of the technology because their motivation puts an emphasis on clear visuals that outline health metrics they are interested in that make it easy to take action based on the interface's feedback of their current state. This might conflict with other stakeholders because while the design might be optimized to be simple and usable, other stakeholders might want to visualize more metrics and data that might conflict with device users' privacy and complexity in the ease of use of the interface.

The motivation for healthcare providers is specifically affecting the design of the technology because their motivation puts an emphasis on long term data collection that can reveal patterns and trends about a user's condition, data accuracy, and medical system integration. This might conflict with other stakeholders because the priority for this design might conflict with a user's motivation for simplicity and privacy because of the constant monitoring and evaluation of data.

The motivation for technology companies might affect the design of the technology because their motivation puts an emphasis on keeping user engagement and retention high so they would design the interface to contain features to do so. This might conflict with other stakeholders' motivation for the design because this might increase the cognitive load of users, unintentionally make them feel a negative emotion towards the interface and healthcare providers who might prefer clean, accurate data over user engagement data.

3 QUESTION 3

3.1 Paper #1: StoryMap – A Fitness Tracking Tool for Low-SES Families

The first paper that caught my interest was StoryMap: Using Social Modeling and Self-Modeling to Support Physical Activity Among Families of Low-SES Backgrounds by Saksono et al. This paper discusses how fitness tracking applications can positively impact the physical activity of families with low socioeconomic statuses. To discover this, the authors conducted a 5-week study on 16 low-SES families using StoryMap, a fitness tracking app they created and designed for allowing family members to see and reflect on their own activity as well as their family members activity. This app was designed around two models of social cognitive theory, social modeling[1] and self-modeling[2]. In this study, the authors observed the families' interactions with the app and interviewed them to determine how their attitudes towards physical activity changed over time using the app. They discovered that the social modeling aspect helped motivate family members to participate and the self-modeling aspect of the app allowed them to feel more confident in being able to be more active which in turn can help improve their health behaviors. Based on these findings, design principles for social personal informatics tools are given by emphasizing the incorporation of social and self-modeling to improve the impact in users' health behavior in using these tools. I found this paper particularly interesting because as someone who uses a fitness tracking app daily and is passionate about bodybuilding and

[1] learning from other people's behavior
[2] self-reflection of one's behavior

health, I found how social interaction through technology can impact one's health insightful.

3.3 StoryMap & HCI – Distributed Cognition, Mental Models, and Feedback

The ability to share actions across users is a direct representation of using the concept and lesson on distributed cognition because it allows for not only an individual users cognitive interpretation of their actions, but their family members interpretation of their actions. This is also the goal of the social modeling theory which synchronizes with distributed cognition well where these cognitive processes of analyzing and reflection are shared across the family members using the interface rather than being done by just the individual (Saksono et al., 2020, p. 1).

The lesson on mental models and representations discusses how representations are formed through the interface in which StoryMap demonstrates users doing by mapping the actions they take to their current health status, ultimately allowing them to make adjustments to their actions to reach their health goals. (Saksono et al., 2020, p. **4**).

Lastly, another concept I saw a connection to in this paper to are constant feedback cycles that the mapping (described above) enables users to evaluate their actions, make changes or continue what they are doing to ensure they reach their goals (Saksono et al., 2020, p. **4**). The emphasis on clear feedback and usability is applied here because users not only visualize their behavior directly with the interface, but indirectly as well through the application of social modeling where users get feedback on their actions from their family members.

3.4 Paper #2: The Design Space of Wearables for Sports and Fitness Practices

The second paper that caught my interest was The Design Space of Wearables for Sports and Fitness Practices by Vidal et al. This paper discusses the current impact of wearable devices in sports and fitness and how their design can be improved to better support it's users' goals beyond data tracking. To do this, the authors surveyed 47 prior research publications regarding wearable devices for sports and fitness and conducted a thematic analysis identifying trends and patterns in how these wearable devices support their users and concluded with a design space for future wearable across 4 categories. I found this paper interesting because I enjoy optimizing my health and fitness through wearable devices it resonated with my experience with wearable devices.

3.5 Wearables & HCI – Distributed Cognition, Qualitative Evaluation, and the Gulf of Evaluation

The first connection I made with this paper was that of distributed cognition because it involves coaches and teammates that distribute the cognitive load of interpreting the data from the wearable devices and providing feedback to the primary user (Vidal et al., 2020, p. 5). This relates to the lesson on distributed cognition and how and how the cognitive load can be shared across an artifact.

The second connection I made with this paper was with the concept of qualitative evaluation because it used 47 existing research papers on wearables regarding sports and fitness and conducted a thematic analysis, a form of qualitative evaluation, to determine trends and patterns that could be further improved to optimize the design space for the wearable device users (Vidal et al., 2020, p. 2-4). This relates to the lesson on qualitative evaluation and how insights can be drawn through themes, trends, patterns, and other non-quantitative data.

The third connection I made with this paper was with the concept of the gulf of evaluation in the lesson on feedback cycles because their solution, an improved design space, is motivated by the fact that the current gulf of evaluation of wearable devices can be improved. It states that they are helpful in presenting a lot of data, but interpreting and connecting it to real-world situations isn't super helpful, ultimately creating a gap in the bridge of evaluation between feedback and how to evaluate it (Vidal et al., 2020, p. 1). This relates to the less on the gulf of evaluation because the design space the researchers propose helps narrow that bridge.

4 QUESTION 4

4.1 LLM's and Your Diet!

The first conference I chose was ACM Conversational User Interfaces and selected the paper Exploring Multi-LLM Collaboration to Power Conversational Recommender System: A Case Study of Dietary Recommendation by Minhui Liang and Yuhan Luo which explores multi-LLM collaboration[3] to improve a

[3] Where multiple LLM's are used to divide a main, complex task into sub-tasks

conversational recommender system (CRS) that delivers personalized dietary recommendations based on age, gender, taste preferences, and social lives of the user. To do this, the authors developed *SmartEats* to compare single and multi-LLM's in the conversation and recommendation phases in which they conducted an online experiment comparing 4 different configuration versions to evaluate performance and user experience. They concluded that versions with multi-LLM's had better adaptability to conversations, greater user experience, and more precise recommendations making it easier for the users to use compared to single-LLMs (Liang & Luo, 2025, P1). I found this paper interesting because as someone interested in health and AI, I found it insightful how we can apply similar software principles like divide-and-conquer to LLM's to optimize user experience.

4.2 Multi-LLM's & HCI – Mental Models, ANOVA, and Cognitive Load

The first connection I made with this paper was that of mental models because the design of SmartEats maintains a very high rate of conversational coherence with the user by adapting to questions based on the context ultimately allowing users to predict that by speaking to the conversational manager, they will get an accurate response based on their context back (Liang & Luo, 2025, P1). This also relates to the concept of synthesizability in mental models because not only can users predict the outcome of their actions, they can view a clear mapping between the context they provide, and the recommendations given to them.

The second connection I made with this paper was with the concept of evaluation, specifically using the quantitative evaluation method of ANOVA to compare the outcomes across 4 different configurations of single vs multi-LLM manager and recommendation engines (Liang & Luo, 2025, P4). This experiment was consistent with what was taught in the lecture because it compares the average outcomes of the 4 different configurations as categories and measured them across participant perceptions of recommendations, nutritional balance, and similarity scores of recommended dishes (Liang & Luo, 2025, P5).

The third connection I made with this paper was with the concept of cognitive load in the lesson on human abilities because the use of multi-LLMs to break the complexity of the CRS into multiple tasks behind the scenes allows for reducing the demand of a single LLM and dividing it onto multiple. As a result, the more context-aware multi-LLM CRS does better at adapting to questions and generating recommendations ultimately reducing the mental effort needed to use the interface and easier for users to communicate with the system.

4.3 Drifting on Ice! Emulated…

The second conference I chose was ACM Automotive User Interfaces and selected the paper Gliding on Simulated Ice: Effect of Low-μ Emulation on Drift-Training by Yasuda et al. This paper explores whether simulating low-μ (low friction) drifting in a real vehicle can help novices learn to drift safer and more effectively than normal friction conditions by conducting a between-group study to compare the groups on performance, motivation, and workload (Yasuda et al., 2024, P1). I found this study interesting because as someone who is passionate about cars and who knows how to drift, this would have been really useful to have been able to use when I first learned!

4.4 Drifting & HCI – Cognitive Task Analysis, Human Abilities, and

The first connection I made with this paper was that of cognitive task analysis because the researchers designed drifting in a low-friction environment to slow the task down and give drivers more time to process the multiple steps that come into play when drifting (Yasuda et al., 2024, P4). This allows them to better understand and perform the task by considering a user's complex perception and decision making.

The second connection I made with this paper was with the concept of human abilities and how the researchers considered how learning to drift can be optimized by appealing to the multiple human abilities that are used. By simulating low-friction conditions, users' cognitive load is decreased, and their focus can be directed to processing sensory information as well as store the necessary actions in their short and long-term memory to drift successfully (Yasuda et al., 2024, P5-6).

The third connection I made with this paper was with the concept of research methodologies and utilizing quantitative data. This study was a good example of using numeric data from ANOVA and t-tests on two training groups (different levels of low-friction) and comparing the average performance and user experience with interval and ratio data (Yasuda et al., 2024, P2-4).

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6 APPENDICES

Transcript of AI-assisted product naming Brainstorm (ChatGPT)

Date: february 2026

Ai system: chatgpt (openai large language model)

Purpose: brainstorming brand names for an ai-driven bodybuilding and fitness application.

Transcript

User:

Hello, i am brainstorming product names for an application i've created. The summary of the application is the following: it's a health fitness app dedicated to helping users achieve their fitness goals. This came from a passion of bodybuilding and sculpting a fit and good looking physique that makes me feel confident, healthy, and the best version of myself. I'd like you to brainstorm names that revolve around what the app does and the motivation behind it. It needs to be something concise, easy to say, and rolls off the tongue.

Assistant:

Nice — this is a strong concept and you already have a clear emotional core: confidence, physique, discipline, becoming your best version. I'll keep everything short, brandable, and easy to say. I grouped ideas by vibe so you can see directions.

Physique/aesthetic-focused names: physiq, sculpt, forme, shredly, aesth, peakform, built, forgefit, ironform.

Best-version/confidence names: ascend, evolve, primeyou, refined, betteru.

Bodybuilding culture names: mass, bulkcut, recomp, ironpath, pump.

Ai-leaning names: fitai, formai, coachai, fitlogic, repsync.

Premium brand-ready names: striv, atlas, core, zenith, pulse, vigor.

My top picks: physiq, sculpt, peakform, forgefit, evolv, trueform, core, atlas.

Quick questions:

Hardcore bodybuilding or mainstream fitness?

Tech/ai or lifestyle brand?

One word or two?

Premium or playful tone?

Future company brand?

User:

It should feel tech/ai, one word, clean, and it is something that can turn into a brand/company later on.

Assistant:

Perfect — that gives a clear lane: tech/ai feel, one word, clean, long-term brandable.

Top overall brand names:

Physiq, forma, evolv, aesth, sculpt, prime, vanta, striv, forge, core.

Ai-leaning:

Optiq, axon, neura, quant, modus, praxis, kinetiq.

Physique + tech fusion:

Morphiq, aestiq, formiq, ironiq, volum, hypertro.

Top recommendations:

Physiq, forma, evolv, aesth, striv, forge, core, vanta.

Quick check:

Do you want it more like strava or more like openai? Hardcore bodybuilding or clean tech startup?

User:

I like physiq, any other suggestions like that? It should be more bodybuilding brand like.

Assistant:

Got it — bodybuilding-first, clean, one word, brandable like physiq.

Closest to physiq:

Aestiq, formiq, morphiq, kinetiq, ironiq, athletiq, sculptiq, massiq, frameiq.

Premium bodybuilding brand names:

Sculpt, forge, anvil, frame, titan, atlas, prime, peak, iron, hypertro.

Modern gym-brandable:

Striv, forma, core, vigor, valor, zenith, modus.

Top picks if you like physiq:

Aestiq, formiq, sculptiq, kinetiq, ironiq, hypertro, volum, morphiq.

Honest take: physiq is already one of the strongest possible names.

Quick questions:

Do you want the name to clearly imply ai or just feel modern?

Do you plan to build a clothing or supplement brand later?

User:

I'll go with physiq.ai — thank you for the help!

Assistant:

Great choice — physiq.ai is genuinely strong.

It's bodybuilding-centric, clean, memorable, and scalable into a full company.

Brand positioning ideas:

"ai-driven physique optimization."

"the operating system for your physique."

Next steps:

Secure domains (physiq.ai, physiqapp.com).

Secure social handles.

Develop logo direction and brand identity.

Transcript of Fitness App Feasibility Feedback (ChatGPT):

User:

I am developing a fitness app that seeks to use computer vision to analyze optimizing people's fitness goals whether it's losing fat, building muscle, or both simultaneously. It will utilize the most accurate AI models to determine this and aims to be more accurate than a dexa scan. Can you give me feedback on if this is a feasible idea or not and if there are other tools like this on the market that I would be competing against?

Assistant:

You’re thinking in a very hot and technically plausible direction — but the claim “more accurate than a DEXA scan” is where feasibility gets tricky. I’ll break this down honestly from a product + AI standpoint.

Short answer:

- Yes, the idea is feasible.
- Beating DEXA accuracy with only images is extremely difficult.
- You can build something valuable and competitive without that claim.
- If executed right, this could be a strong startup — especially given a background in software engineering and bodybuilding.

Technical feasibility:

Computer vision can estimate body composition with reasonable accuracy. However, truly beating DEXA with photos alone is not realistic today. Photos capture external geometry and lighting; DEXA measures internal tissue composition via X-ray. Even top research models struggle to match DEXA accuracy.

However, strong performance can be achieved using:

- multi-angle photos
- weight history
- standardized poses
- trend tracking over time

The optimization recommendations component (bulk, cut, recomposition guidance) is highly feasible and potentially more valuable than perfect body-fat estimation.

Competitors include:

- Spren
- Zozofit
- Fit3D
- InBody ecosystem
- general fitness trackers (MyFitnessPal, etc.)

Most competitors either:

- estimate body composition

- track nutrition
- or provide coaching

Few combine all three into a unified optimization system.

Major technical challenges:

- dataset acquisition (images paired with DEXA scans)
- lighting/pose variability
- distinguishing muscle vs fat visually

Key opportunity:

Instead of positioning as a "DEXA killer," position as an AI physique optimization coach that helps users make better decisions weekly.

Focus on:

- trends over time
- weekly adjustments
- actionable recommendations

Potential roadmap:

- Phase 1: weight tracking and recommendations
- Phase 2: photo logging
- Phase 3: body composition estimation
- Phase 4: full AI coaching

Summary:

The strongest version of this idea is an AI-powered physique optimization system rather than a pure scanning tool.

User:

Can you give me a summary of what the goal is from these conclusions of what's feasible and to be unique compared to other apps?

Assistant:

Core goal:

Build the most actionable AI physique-optimization app — not simply the most accurate body-fat scanner.

The winning positioning is:

Help users make the best decisions to lose fat, gain muscle, or recomposition using computer vision and data trends.

Feasible elements:

- body composition estimation from photos (approximate)
- tracking trends over time
- providing optimization recommendations

Not realistically feasible:

- consistently beating DEXA accuracy using only images
- perfect body-fat estimation from a single photo

Differentiation:

- focus on decision-making rather than measurement
- track progress over time
- provide weekly coaching adjustments
- target serious lifters

Unique positioning:

An AI physique intelligence platform combining:

- computer vision
- progress tracking
- predictive modeling
- weekly recommendations

Mission statement:

Help lifters make better physique decisions using computer vision and AI-driven coaching.

User:

Can you provide a concise summary so I can use it to design the app’s interfaces?

Assistant:

App design goal:

Build an AI-powered physique optimization app that helps users make better weekly decisions using photos, weight data, and progress trends.

The app functions as a digital physique coach.

Core user outcome:

Users quickly understand:

- what is happening to their body
- whether they are progressing
- what to do next

Core loop:

- Log weight and photos
- Analyze trend
- Evaluate progress vs goal
- Provide adjustment
- Repeat weekly

Key design principles:

- emphasize trends over exact body-fat numbers
- decision-first UI
- built for serious lifters
- minimal and performance-focused

Core screens:

- dashboard
- weekly check-in
- physique analysis
- adjustment recommendations
- projections

Primary question each screen should answer:

- Where am I now?
- Am I on track?
- What should I change?
- Where will I be soon?

Product definition:

A weekly AI check-in app that analyzes physique progress and recommends adjustments to reach goals faster.

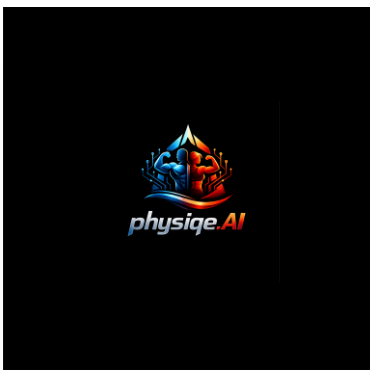
Transcript of Logo Design Request (ChatGPT)

User:

Can you make the design for the logo/brand for my app? The summary of the application is the following: it's a health fitness app dedicated to helping users achieve their fitness goals. This came from a passion of bodybuilding and sculpting a fit and good looking physique that makes me feel confident, healthy, and the best version of myself. Its name is physiqe.AI.

Assistant:

(Generated initial logo concept based on description.)



The logo design featured a strong fitness-technology aesthetic, incorporating muscular silhouettes and futuristic elements to reflect both bodybuilding and AI-driven optimization.

First Design Iteration Feedback

User:

Can you do just the man doing the front double bicep and all blue instead of the way it's currently pictured? I think it would be a cleaner image. Also change the physiqe.ai to be a gradient going from dark blue to white instead of what it currently is.

Assistant:

(Generated revised logo concept.)

Updated design included:

- Single male silhouette in a front double-bicep pose
- Entirely blue color scheme for a cleaner identity
- Dark-blue-to-white gradient typography for physiqe.AI
- Simplified futuristic fitness-AI branding